

RESEARCH-EVIDENCE — the consolidated record (WP5.7)

Generated by `scripts/build_research_evidence.py` ; every number below is read from a committed artifact, and re-running the script reproduces this file byte-identically. Negative results are sections, not footnotes.

The seals this record is judged under: `pre-registration.lock`

`sha256:bf0b6d8936975a09f8c92f8ec8abc24aa43b1c18e26d5cd32dbf2fb9bd21b58d` · G3

`sha256:938aef3ee9c2f85b0dd7856ae89e6122c0058337c65170c0f05fa01753ba51b8` · G5

`sha256:601e3f826b089de171783eb397b6697ec2d5a967a606659a303ed47ba79ea615` .

The standing caveat, verbatim at the top because it governs everything below: *a policy that wins in the ensemble has been tested against the policy AND the generator together -- say so plainly.*

RQ1 — Fidelity: does the generator reproduce the world's structure?

- **The sealed battery record** (`G2-EVIDENCE.md` , `ABLATION.md` ,

`ablation.json`): the hierarchical generator was selected by the sealed multi-seed rule over the bootstrap benchmark; the ablation grid's verdict is the committed record, generated not hand-assembled.

- **Campaign-2 promotion** (`artifacts/campaign2/PROMOTION.md`): the

fifteen-factor generator (FX, CAPE, revived HY) beats `bootstrap-v1` in EVERY seed on both sealed clauses, both sides passing their batteries clean:

seed	challenger elicit.	benchmark elicit.	d	band exceed. (C vs B)	beats
0	-2.5591	-2.2131	-0.3461	12 vs 13	yes
1	-2.5163	-2.2132	-0.3031	5 vs 11	yes
2	-2.5116	-2.2139	-0.2978	8 vs 12	yes

- **Accepted limits, owner decision C3** (mechanism trace:

`Instructions/campaign2-regime-fix-options.md`): level-factor half-lives revert at the block-reassembly timescale; equity/hml lost-decade excess (part start-state artifact). The 10yr tier claims NO pass (`G2-`

RQ2 — Held-out regimes: does it generalize to data it never saw?

- **The severe test is INCONCLUSIVE, and that is the finding**

(artifacts/wp211/SEVERE-TEST.md): excluding the 1970s and regenerating from the 1965 state moves the era-frequency gap by ~6-8% of a gap that exists with the decade in-sample. The reading is a recorded human judgement (S2-SEVERE-GATING); the L3 leg is structurally vacuous (0 blocks dropped) — stated in the evidence, not smoothed.

- **THE ONE-SHOT HOLDOUT IS SPENT** (owner authorization verbatim "Go",

2026-08-03; spec sealed AM-2026-08-02-006; one generation at base_seed 20260000, one token read, one table -- artifacts/wp56/holdout-evaluation.json). The result, verbatim both ways:

- **Primary (drawdown surprise): -0.3952** -- realized max

drawdown 0.248 (the 2022 bear) against the ensemble's p95 warning 0.644. Reality stayed inside the warning: one sentence of support for decade-scale usability, with the single-realization caveat stated (n=1 decade; confirmation, not proof).

- ****Realized terminal wealth sits at the 99.6%**

percentile of the ensemble** -- the cone contained the downside and badly under-spanned the realized upside. Reported as a finding, no footnote.

- **cpi band coverage 0.000**: the realized 2021-2023 inflation burst

escaped the bands every month -- the C3-accepted inflation-era limitation (Instructions/campaign2-regime-fix-options.md) manifesting where its documentation predicted. cape_v's zero is partly definitional (holdout- span vs train+val demeaning centers). fx_usd went UNREAD (a reader fallback bug, logged in the artifact); per the sealed spec nothing re-runs -- the gap publishes as a gap.

factor	realized months	coverage in [p5, p95]
cape_v	45	0.000
cpi	65	0.000
equity_mkt	65	0.785
equity_vol	67	0.761
funding_spread	12	0.000
hml	65	0.708
hqm_curve	66	0.682

hy_spread	67	0.985
ig_spread	66	0.803
mom	65	0.908
policy_rate	66	1.000
smb	65	0.785
ust_10y	67	0.776
ust_2y	67	0.836

RQ3 — Decision value: do ensemble-informed decisions beat benchmarks?

- **The final walk-forward ran on the sealed protocol**

(artifacts/wp57/walkforward-final.json ; folds per AM-2026-08-02-005, five of six benchmark policies — static_endowment_mix is UNCOMPUTABLE, commodities unsourced, RFR-8). Wilcoxon vs static_60_40 on terminal log wealth (effect sizes always, per the sealed rule):

policy	statistic	p	rank-biserial
history_only_optimization	5.0	0.0195	-0.818
gaussian_monte_carlo	5.0	0.0195	-0.818
bootstrap_ensemble	5.0	0.0195	-0.818
fixed_heuristic_rules	20.0	0.4922	-0.273

- **The frozen primary — drawdown surprise — is NEGATIVE in all ten folds**

(mean -0.1863, worst -0.0709): realized drawdowns never exceeded the promoted generator's p95 warning. Read BOTH ways: the cone never under-warned, and it is loose against a calm realized decade (unconditional year-one cone vs 2011-2020). Model-level look-ahead stated: the generator trained on 1990-2020; a per-fold-retrained variant is named future work.

fold	drawdown surprise
1	-0.0965
2	-0.2035
3	-0.2460
4	-0.2399
5	-0.1841
6	-0.2154

7	-0.2731
8	-0.1277
9	-0.2060
10	-0.0709

- **Decision-density pilot** (`artifacts/wp55/density-pilot.md`): across 300

worlds, 43% of total |contribution| sits in the first two decision windows ("consequence concentrates at to", supported at pilot scale) with a mid-decade bump above the time-decay line; the window-vs-regime confound of fixed preset schedules is stated, and the generated-world full run (budgeted from measured rates in the same report) is what breaks it.

- **The counterfactual machinery is exact** (`ah.gen.recone` ,

`ah.eval.counterfactual`): re-coning conditions every layer exactly (the wp5-03 verification record), and the was-it-a-good-call metric scores paired deltas on identical continuations.

RQ5 — Commitment behaviour under pressure

Harness only, by design, until players exist (the kickoff's phrasing). The pacing levers, forced-sale waterfall, policy twin (DN-5 v0.2) and the tournament's cohort export are built and tested; behavioural claims await consented human runs (DN-6's M4 consent clause is the blocking item, on the counsel track).

RQ4 — Practitioner robustness panel

PARKED, owner decision D-K5-2 ("until the product is polished — and wanted then"). Re-raised at the beta/M4 boundary; recruitment is not the constraint.

The negative results, collected

1. **G1's 2022 replay: FAIL** (`WP3.11`) — the honest answer to "does the engine reproduce 2022?" with the mark-lag limitation recorded. 2. **The severe test: INCONCLUSIVE** (RQ2 above) — reported as such. 3. **B' and A' both null** (`Instructions/campaign2-regime-fix-options.md`): guidance moves persistence the wrong way; the residual parameterization moves nothing — the diagnosis that followed relocated the defect and the chased anchors themselves. 4. **The 10yr tier claims no pass**; its two structural-gap regime metrics remain NaN-marked. 5. **static_endowment_mix and two D4 strategies are uncomputable** (commodities unsourced) — reported wherever they appear.

Reproducibility appendix

- Every RunRecord regenerates its outputs from stored inputs

(`ah.store.runrecords.verify_run`); `ah inspect RUN_ID` renders any run's figure page by REGENERATING it, digest-verified on every render.

- Every committed evidence artifact names its generating script; the

ablation tables, promotion verdict, density pilot and walk-forward final are all script outputs.

- The three locks above hash the thresholds AND the judging code together;

amendments are append-only (`governance/amendment-log.yaml`).

Not investment advice.